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Teaching and Learning Statement

I strive to emphasize three tenets in my teaching: communication, understanding, and application. Clear communication leads to better understanding, and better understanding helps in successful application of knowledge. Regardless of the class, whether it be large lecture or recitation, I believe that these tenets provide a supportive and productive learning environment for students.

Teaching Experience

At Auburn University, I was the instructor of record for two classes and a recitation leader for five classes. I regularly participated in pedagogy seminars and strove to constantly improve my teaching for my students. I was recognized for my efforts in the classroom by receiving the college's Outstanding Graduate Teaching Assistant award: a competitive award granted every year to three graduate students across the five departments comprising the College of Sciences and Mathematics.

Now at Iowa State University, I have taught in nearly every aspect of the classroom, ranging from large lectures (> 250 students) to medium classes (~ 60 students) to smaller recitations (~ 30 students). I participate in our Coalition for Excellence in Undergraduate Mathematics Education (CEUME) teaching seminar to stay up to date on current best practices for teaching and learning in the classroom.

I also led a group of undergraduate students on a research project related to combinatorial commutative algebra with simplicial complexes and Stanley-Reisner theory as part of the Iowa State Mathematics Research Teams (ISMaRT) program. At the end of the semester, the students shared their progress via a poster presentation to graduate students, postdocs, and faculty at a department event.

Developing Understanding with a View toward Applications

I view the classroom as an opportunity for students to get hands-on, practical experience with the math they learn. Through interaction with me and their peers, I want students to understand the math is a collaborative endeavor, and that they are learning how to create and apply general frameworks to problems they will encounter.

At Iowa State, I have had the pleasure to teach linear algebra to a class of about 60 students. Different from a large lecture and a comparatively smaller recitation, determining what topics and activities were designated as "traditional lecture" versus "active learning" was challenging. Here, the class size was manageable enough so that I could walk around and help groups with the content, but without a separate lecture component, the class could not *just* practice linear algebra problems.

I opted to give students collaborative warm-up problems at the start of class, focusing on concepts from the past class. Then, I would alternate between introducing new concepts through definitions, examples, and theorems, to having the class split into groups to practice and extend the content that was just introduced. This seemed to work well as students who were feeling comfortable with the content could help their peers—teaching is an effective way to reinforce learning—whereas those less comfortable had dedicated time during class to ask questions and seek understanding.

As the semester progressed, we developed more mathematical tools to use in understanding a linear transformation. Once eigenvectors and eigenvalues rolled around, I then created a software demonstration in Python for my students to exhibit the geometry of linear transformations. The course culminated in singular value decomposition, my favorite application of linear algebra. To help showcase the power of all of the linear algebra machinery we learned throughout the semester, I created multiple demonstrations applying singular value decomposition to real problems; for instance, problems in the field of computer vision such as*principal component analysis* and *eigenfaces*.

Course Design and Analysis

Joint with Lanius and Gilroy, we developed the Course Alignment Analysis Tool (CAAT) to mathematically analyze the alignment of course objectives in classes and published our findings in the International Journal of Mathematical Education in Science and Technology. The CAAT has the instructor rank which learning objectives should be emphasized the most (ties are allowed), and then has them count how many times each objective appeared in an assignment, thus generating an observed ranking of objectives. The discrepancy or distance in these rankings is calculated and used to quantify how well an assignment aligns with the expectations of the instructor: the smaller the distance, the better the alignment (see Figure 1).

Alignment summaries are generated based on the quartiles of the distribution of distances. The first quartile (green) indicates an "Excellent" alignment, the second (yellow) an "Acceptable" alignment, and the last two (red) a "Poor" alignment.

When analyzing the impact of the CAAT on the perception of uniform homework, we found that the CAAT affected participants' definitions of homework quality. Especially with novice instructors, the CAAT shows promise as a professional development tool as it quantifies the alignment between proposed and assessed learning objectives.

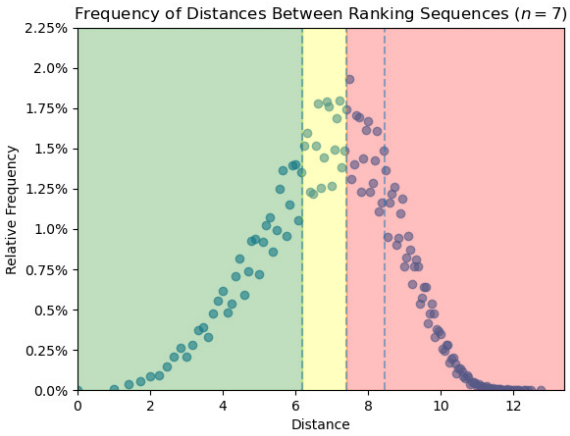


Figure 1: The distribution of distances on $n = 7$ learning objectives.

Now at Iowa State, I have been working with course coordinators in implementing Gradescope for large, coordinated courses such as the calculus sequences and linear algebra. Tools such as Gradescope offer functionality including anonymous and asynchronous grading, as well as rubric visibility for students. All of these incorporate into having more rigorous and transparent grading that benefits instructors, through backward design of rubrics, and students, through clear expectations in grading standards.

Closing Remarks

In all of these contexts, I try to make teaching and mentorship an inviting and personal endeavor for all parties involved. My approach to teaching is continually evolving by incorporating research-backed methodologies and feedback from my students to tailor it to the needs of the classroom; having this feedback cycle encourages open communication between me and my students. I constantly remind my students that this is really *their* course and that I am there to facilitate and reinforce their learning.